CO2 - AT1 - Analytical Problem solving

① Construct an efficient system using two stacks within a single array. Compare fixed partitioning and dynamic allocation strategies.

Ans: A two stack implementation in a single array improves memory utilization by allowing both stacks to share the same array.

- stack 1 grows from left to right.
- stack 2 grows from right to left.
- Overflow occurs only when top1 + 1 = top 2

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Fixed Partitioning

- The array is divided into two equal parts
- Each stack has a fixed size.
- If one stack becomes full, it overflow even if the other half is empty.

Dynamic Allocation

- Both stacks share the entire array.
- A stack can use extra space if the other stack has free space.
- Overflow occurs only when the entire array is full.

Comparision

Feature	Fixed Partitioning	Dynamic Allocation
Memory Utilization	Low	High
Overflow	May occurs early	Only when array is full
Space Effiency	Poor	Excellent
Time Complexity	Push / Pop : O(1)	Push / Pop : O(1)
Best for	Predictable workloads	Variable workloads.

Conclusion

Dynamic allocation is more efficient because it reduces unused space and delay overflow, making it suitable for varying workloads.

② Queue based ticketing system with VIP customers (Priority Queue)

Ans A simple queue serves customers in First-In-First-Out (FIFO) order. Every customer is treated equally.

A priority queue allows VIP customers to be served before normal customers based on priority.

<u>Simple Queue</u>

- Follows FIFO
- Fair to all customers.
- Waiting time depends only on arrival order.
- Operation (enqueue / dequeue) take O(1) time.

<u>Priority Queue</u>

- VIP customers are served first
- Reduces waiting time for VIPs
- Normal customers may wait longer (starvation is possible)
- Insert / Delete operation are usually O(log n) using a heap.

<u>Comparison</u>

Features	Simple Queue	Priority Queue
Service Order	FIFO	Priority - based
Fairness	High	Lower for normal customers.
VIP Support	~~Equal~~ No	Yes
Waiting Time	Equal	Less for VIPs
Complexity	O(1)	O(log n)

<u>Conclusion</u>

A simple queue is fair and easy to manage, while a priority queue is better when VIP customers require faster service. The choice depends on whether fairness or priority is more important.